

AUTOMATIC LIGHT AND OVERCHARGING CONTROL With Voltage-Level Indicator

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This circuit can be used for automatic control of lights and 6V battery charging. It monitors the battery level and switches to charging and overcharging protection modes accordingly.

The block diagram is shown in Fig. 1. The circuit has four units: voltage-level indicator, automatic charging and overcharging protection, light control and power supply. When battery voltage is less than the threshold value, the relay energises to connect the battery to the charging source. It also prevents overcharging by constantly monitoring the battery voltage level.

Circuit and working

Circuit diagram for automatic light and overcharging control with voltage-level indicator is shown in Fig. 2. It is built around five 741 operational amplifiers (IC1 through IC4, IC6), an 8V 7808 voltage regulator (IC5), a bridge rectifier (BR1), two 12V single-changeover relays (RL1 and RL2), three 2N2222 transistors (T1 through T3), three 1N4007 rectifier diodes (D1 through D3), a 6V battery (BATT.1) and a few other components.

Voltage-level indicator. Voltage-level indicators are built around IC1 through IC4 to indicate four different battery levels. IC1 through IC4 are used as comparators. These compare four different input voltages (5.5V, 4V, 2.7V and 1.4V) through voltage dividers formed by resistors R1 through R3, R6 and R9. The corresponding voltage levels are indicated using LED1 through LED4, respectively.

Battery charging. Automatic charging and overcharging protection circuit is built around transistors T1

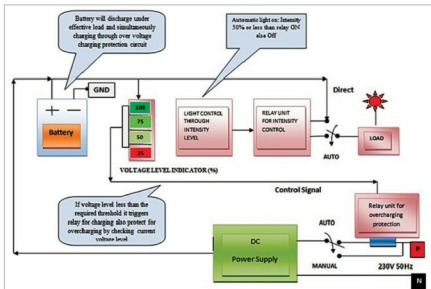


Fig. 1: Block diagram for automatic light and overcharging control with voltage-level indicator

TABLE I
LED INDICATION AND RELAY STATUS FOR DIFFERENT BATTERY VOLTAGE LEVELS

LED On	LED Off	Battery Voltage (BV)	Relay	BV (%)
LED1 through LED4		Above 5.5V	RL1 de-energised, charging 'off'	100
LED2 through LED4	LED 1	Below 5.5V and above 4.0V	RL1 energised, charging 'on'	75
LED3, LED4	LED1, LED2	Below 4.0V and above 2.7V		50
LED4	LED1 through LED3	Below 2.7V and above 1.4V		25
LED5		In dark, the load is 'on' if S2 is in auto mode	RL2 energised, load 'on'	
LED6		Battery is charging	S1 is either in auto or manual mode	

TABLE II
FUNCTIONS OF SWITCHES

Switch	Position	Function
S1	Auto Manual	Automatic charging and overcharging protection Disables auto-charging, so the battery needs to be charged manually
S2	Auto Direct	Load turns on in the dark and turns off in light automatically Disables LDR1; the load needs to be turned on directly